

TRI-CITIES, TN, USA

WMO: 723350

Lat: 36.480N

Lon: 82.399W

Elev: 456

StdP: 95.96

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13877

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-8.1	-17.2	0.9	-7.0	-14.6	1.1	-5.4	10.4	5.7	9.1	4.3	1.9	250	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.5	22.2	31.3	21.9	30.3	21.6	23.9	29.6	23.4	28.8	22.8	28.0	2.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.0	26.0	21.8	17.4	25.4	21.3	16.8	24.9	73.9	29.7	71.7	28.9	69.5	27.9	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.0	-14.9	34.1	3.9	1.8	-17.7	35.4	-19.9	36.4	-22.1	37.4	-24.9	38.7		
(5)			-15.5	25.0	3.7	0.6	-18.2	25.4	-20.3	25.8	-22.4	26.1	-25.2	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	2.2	4.5	8.6	13.7	18.3	22.4	24.0	23.6	20.4	14.1	8.0	4.2		
	DBStd	8.80	5.97	5.50	5.13	4.41	3.68	2.46	2.07	2.09	3.20	4.44	4.71	5.17		
	HDD10.0	820	250	167	88	17	1	0	0	0	0	14	92	191		
	HDD18.3	2295	499	388	302	146	48	3	0	1	17	142	311	437		
	CDD10.0	2175	9	12	44	128	257	372	433	422	311	142	32	12		
	CDD18.3	608	0	0	1	8	46	126	175	164	78	11	1	0		
	CDH23.3	5128	0	3	13	123	456	1035	1394	1311	664	123	6	0		
	CDH26.7	1693	0	0	1	16	102	345	515	480	214	21	0	0		
(14) Wind	WSAvg	1.9	2.3	2.4	2.6	2.4	1.9	1.6	1.5	1.2	1.3	1.5	1.7	1.9		
(15) Precipitation	PrecAvg	1031	82	87	98	84	96	95	117	82	76	62	73	88		
	PrecMax	1315	135	160	234	179	220	187	232	180	180	135	171	172		
	PrecMin	779	35	19	33	5	33	19	45	14	13	0	13	5		
	PrecStd	132	28	37	45	39	44	41	48	32	43	35	33	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.5	22.4	25.1	28.6	30.8	33.1	34.1	34.0	32.7	28.9	23.9	20.9		
		MCWB	14.4	15.4	15.8	17.9	20.1	21.8	23.0	22.8	20.6	20.2	16.3	15.3		
	2%	DB	16.5	18.5	22.6	26.5	29.1	31.5	32.3	32.4	31.0	26.4	21.1	17.1		
		MCWB	12.7	12.4	14.5	16.7	19.7	21.6	22.8	22.4	20.8	18.1	14.4	12.9		
	5%	DB	13.9	15.7	20.2	24.7	27.7	30.2	31.1	31.0	29.3	24.3	18.9	14.6		
		MCWB	10.7	10.7	13.0	15.9	19.2	21.3	22.5	22.1	20.4	16.8	13.4	11.6		
	10%	DB	11.5	13.2	17.6	22.6	26.2	28.8	29.7	29.5	27.4	22.3	16.6	12.6		
		MCWB	9.1	9.3	11.7	15.1	18.6	20.9	22.2	21.7	19.7	16.3	12.1	10.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	16.7	16.9	19.9	22.4	24.1	25.1	24.5	23.4	22.0	18.2	16.2		
		MCDB	17.6	20.8	23.0	26.0	27.6	29.5	30.5	30.6	28.9	26.6	21.5	19.3		
	2%	WB	13.3	14.0	15.6	18.3	21.3	23.2	24.1	23.8	22.3	20.0	16.2	14.0		
		MCDB	15.9	17.3	20.7	24.2	26.7	28.9	29.7	29.6	27.5	23.8	19.6	15.6		
	5%	WB	11.6	11.9	14.3	17.1	20.4	22.5	23.5	23.2	21.5	18.6	14.6	12.4		
		MCDB	13.4	14.7	18.6	22.4	25.7	27.8	28.8	28.7	26.2	22.5	17.7	14.2		
	10%	WB	9.4	10.1	12.9	15.9	19.5	21.8	22.9	22.6	20.7	17.3	12.8	10.4		
		MCDB	11.2	12.4	16.5	20.6	24.4	26.7	27.9	27.6	25.1	21.0	16.0	12.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	11.0	12.5	13.4	12.7	11.9	11.0	11.7	12.5	13.2	12.6	10.5		
		MCDBR	13.5	15.2	16.3	16.5	14.9	14.0	13.0	13.6	15.1	15.9	15.7	13.3		
		MCWBR	9.5	9.8	9.1	8.1	6.4	5.3	4.7	4.9	5.8	7.9	9.4	9.3		
		MCWBR	11.4	12.2	13.1	13.2	12.6	11.6	11.1	11.4	11.6	12.0	12.0	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.309	0.334	0.375	0.418	0.446	0.481	0.483	0.382	0.352	0.314	0.294		
		taud	2.517	2.486	2.452	2.349	2.239	2.193	2.128	2.099	2.368	2.447	2.520	2.557		
	Ebn at Noon		903	932	934	907	868	840	807	796	871	869	876	886		
		Edh at Noon	79	92	105	123	139	146	154	156	113	95	79	72		
(44) All-Sky Solar Radiation	RadAvg		2.22	2.80	4.01	5.13	5.67	6.11	5.68	5.39	4.61	3.59	2.64	1.95		
	RadStd		0.17	0.35	0.33	0.39	0.41	0.37	0.37	0.24	0.38	0.42	0.28	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (37 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-120	+150	+84
		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+143	+88

Nomenclature: See separate page